Software Engineer Preliminary Assignment

Overview

The manufacturing process shall run through a defined sequence of working steps (for example, Step A → Step B → Step C). Each step may handle one or more orders concurrently, and each step consumes resources such as materials, machine time, or labour.

Order manufacturing progress shall be traceable, which limits visibility into the state of each order, and the consumption of resources.

Objective

You are requested to develop a prototype of a manufacturing process tracking platform that lets users track orders through a multi-step sequence and monitor resource consumption.

Scope

At a minimum, a user of the prototype should be able to:

- create orders and advance them through the defined steps;
- view the orders currently in progress at each step;
- observe resource consumption as work progresses; and
- obtain an overview of the current status of all orders.

The definition of steps, orders, and resources, the rules governing their interaction, and the design of the user interface are available for your creation. The assessment is concerned primarily with your reasoning about the process as a whole rather than with any individual feature.

Due to time constraints, Please manage your time effectively to construct and deliver the platform to the highest achievable quality in both coding and user journey aspects.

Deliverables

- The prototype, provided either as source code in a repository that can be executed locally (with setup instructions) or as a hosted application.
- A written summary of approximately one page, addressing:
    - Summary of the requirements;
    - How do you convert the requirement into the functional specifications;
    - Brief description of software design;
    - Simplified user manual; and
    - Summary of the software.

Conditions

- The choice of programming language, framework, and technology stack is unrestricted.
- You may complete the assignment at their own pace within a period of one week from receipt date.
- Use of AI-assisted tools is permitted; you shall be asked to describe how such tools were used.

Confidentiality

All materials submitted will be used solely for the purpose of this recruitment process and will not be used for any commercial purpose.

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